Qi Ding

Ph.D. Candidate in Management Science and Engineering
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Education

Tsinghua University 2023–present
Ph.D. in Management Science and Engineering, Institute of Energy, Environment, and Economy
Tongji University 2018–2023
B.Eng. in Vehicle Engineering, School of Automotive Studies

Research Interests

Transport decarbonization and integrated energy systems; electric vehicles, charging, and power-system interactions; hydrogen and low-carbon fuels; optimization and data-driven analysis for energy and climate policy.

Publications

- **Ding, Q.**, Ren, J., Zhang, S., & Chen, W. (2026). The role of shared autonomous electric vehicles in decarbonizing China's passenger transport sector. *Applied Energy*, 422, 128259. [DOI]
- **Ding, Q.**, & Chen, W. (2026). Retired Battery Supply and Energy Storage Demand in China: A Provincial-Scale Assessment. *Energy Proceedings*, 65. [DOI]
- Chen, W., Zhang, S., Zhang, Q., Ren, J., & **Ding, Q.** (2025). Assessing China's Province-Coordinated Power System Carbon-Neutral Transition Pathway. *Journal of Energy and Climate Change*, 1(1), 1–15. [DOI]

Manuscript Under Review

- **Ding, Q.**, et al. Aligning Retired EV-Battery Supply with Grid-Storage Demand through Interprovincial Coordination in China. Manuscript under review at *Applied Energy*.

Selected Research Experience

- **National Key R&D Program of China — CCUS Technology Assessment.** Conducted techno-economic assessment of carbon capture, utilization, transport, and storage pathways in support of national technology-roadmap research.
- **World Bank Group — Green China Project.** Used the China TIMES 2.0 integrated assessment model to analyze low-carbon power-sector transition pathways and support policy research on green growth.
- **European Commission (DG CLIMA) — COMMITTED Project.** Contributed to multi-model analysis of climate-policy portfolios and global low-carbon transition pathways.
- **Commercial-Vehicle Carbon-Neutrality Roadmap.** Assessed hydrogen, biofuels, and electrification options for clean energy supply in commercial transport.

Methods and Skills

Integrated assessment, optimization, techno-economic analysis, GIS, and data-driven modeling; Python, GAMS, MATLAB, TIMES, and PyPSA; English (CET-6: 641) and German (Band 6: Excellent).

Selected Honors

National Scholarship (2021, 2022); Shanghai Outstanding Graduate (2023).